



HOMEWORK ASSIGNMENT 2

MICROECONOMIC ANALYSIS

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QUESTION V11.10

A consumer has an expected utility function given by $u(w) = \ln w$. He is offered the opportunity to bet on the flip of a coin that has a probability π of coming up heads. If he bets $\$x$, he will have $w + x$ if head comes up and $w - x$ if tails comes up. Solve for the optimal x as a function of π . What is his optimal choice of x when $\pi = 1/2$?

Let's consider a consumer whose preferences over wealth are represented by the von Neumann-Morgenstern utility function $u(w) = \ln w$, we define for $w > 0$.

This utility function is strictly increasing and strictly concave, implying that the consumer is risk-averse. The concavity of the logarithmic function ensures diminishing marginal utility of wealth, which is central to the consumer's behavior under uncertainty.

The consumer begins with the initial wealth w and is offered a gamble based on the flip of a coin. The coin lands heads with probability $p \in (0, 1)$ and tails with probability $1 - p$. If the consumer bets an amount x , his final wealth is:

- $w + x$ with probability p ,
- $w - x$ with probability $1 - p$

The consumer chooses x so as to maximize expected utility. Under expected utility theory, the consumer evaluates the gamble by computing the expected value of utility across states. Expected utility as a function of the bet size x is given by:

$$EU(x) = p \cdot \ln(w + x) + (1 - p) \ln(w - x)$$

The consumer's optimization problem can therefore be written as

$$\max_x p \cdot \ln(w + x) + (1 - p) \ln(w - x)$$

Subject to the implicit constraints $w + x > 0$ and $w - x > 0$, which ensures that utility is well-defined. To find the optimal choice of x , we compute the first derivative of expected utility with respect to x :

$$\frac{dEU}{dx} = p \frac{1}{w+x} - (1 - p) \cdot \frac{1}{w-x}$$

An interior optimum satisfies the first-order condition

$$\frac{dEU}{dx} = 0$$

Which implies

$$\frac{p}{w+x} = \frac{1-p}{w-x}$$

Now I will illustrate the calculations for solving the optimal bet:

$$\frac{p}{w+x} = \frac{1-p}{w-x} \implies p(w-x) = (1-p)(w+x)$$

$$pw - px = (1-p)w + (1-p)x$$

$$pw - px = w - pw + x - px$$

$$pw - w + pw = x - px + px \implies w(2p-1) = x$$

Thus, the consumer's optimal bet as a function of the probability of heads is $x'(p) = w(2p-1)$. The second order condition is satisfied because the second derivative of expected utility is strictly negative for all feasible x , confirming that the solution corresponds to a maximum, and is described by the following function:

$$\frac{d^2 EU}{dx^2} = -\frac{p}{(w+x)^2} - \frac{1-p}{(w-x)^2}$$

Consider now the special case in which the coin is fair, so that $p = 1/2$. Substituting this value into the optimal betting yields:

$$x' = w(2 \cdot \frac{1}{2} - 1) = w(1 - 1) = 0$$

Therefore, when the gamble has zero expected monetary value, the consumer optimally chooses not to bet. This result follows directly from risk aversion: although the expected payoff of the gamble is zero, the concavity of the logarithmic utility function implies that the expected utility of gambling is lower than the utility of retaining certain wealth w .

In summary, a consumer with logarithmic utility chooses a bet size proportional to both initial wealth and the degree to which the probability of winning exceeds one half. While favorable odds induce a positive bet, a fair gamble is optimally rejected. This result illustrates a fundamental implication of expected utility theory: risk-averse individuals decline actuarially fair gambles due to diminishing marginal utility wealth.

QUESTION V14.20

A monopolist maximizes $p(x)x - c(x)$. In order to capture some of the monopoly profits, the government imposes a tax on revenue of an amount t so that the monopolist's objective function becomes $p(x)x - c(x) - tp(x)x$. Initially, the government keeps the revenue from this tax.

a. Does this tax increase or decrease the monopolist's output?

The monopolist chooses output x in order to maximize profits. When the government imposes a proportional tax on revenue at rate t , the monopolist's profit function becomes:

$$\pi(x) = p(x)x - c(x) - tp(x)x = (1 - t)p(x)x - c(x)$$

The key feature of a revenue tax is that it does not depend on costs but instead reduces the fraction of revenue retained by the firm. To determine how this affects output, we derive the monopolist's optimality condition. Differentiating profits with respect to output and applying the product rule, we obtain the following expression:

$$\frac{d\pi}{dx} = (1 - t)\frac{d}{dx}[p(x)x] - c'(x) = (1 - t)[p(x) + p'(x)x] - c'(x)$$

An anterior optimum satisfies

$$(1 - t)[p(x) + p'(x)x] = c'(x) \rightarrow A1$$

The term $p(x) + p'(x)x$ is the monopolist's marginal revenue. Equation A1 shows that the tax multiplies marginal revenue by the factor $1 - t$, while the marginal cost remains unchanged. In the absence of taxation, the monopolist's first-order condition would instead be:

$$p(x) + p'(x)x = c'(x) \rightarrow A2$$

Because $1 - t < 1$, the taxed marginal revenue curve lies strictly below the untaxed marginal revenue curve for all x . Under standard regularity conditions (specially, that marginal revenue is decreasing in output) this downward shift implies that marginal revenue equals marginal cost at a lower level of output.

Therefore, a tax on revenue that is not rebated to consumers unambiguously decreases the monopolist's output.

- b. Now the government decides to award the revenue from this tax to the consumers of the monopolist's product. Each consumer will receive a "rebate" in the amount of the tax collected from his expenditures. The representative consumer who spends px receives a rebate of tpx from the government. Assuming quasilinear utility, derive an expression for the consumer's inverse demand as a function of x and t

Suppose now that the government rebates tax revenue to consumers in proportion to their own expenditures. A representative consumer who spends px receives a rebate equal to tpx . The rebate directly affects the consumer's effective price, and hence demand. We are assuming that preferences are quasilinear:

$$U(x, m) = v(x) + m$$

Where m denotes consumption of a numeric good and income is y . Quasilinearity implies that income effects are absent for the monopolist's good, which will be crucial for the neutrality result obtained later. The consumer's budget constraint reflects both payment and rebate:

$$m = y - px + tpx = y - (1 - t)px$$

Substituting this constraint into the utility function, the consumer chooses x to maximize

$$v(x) + y - (1 - t)px$$

The first-order condition for an interior solution is obtained by differentiating with respect to x :

$$v'(x) - (1 - t)p = 0 \quad \rightarrow \quad v'(x) = (1 - t)p \quad \rightarrow \quad B1$$

Equation B1 equated the marginal utility of consumption to the effective price actually borne by the consumer. Solving explicitly for p gives the inverse demand function:

$$p(x, t) = \frac{v'(x)}{1 - t}$$

This expression shows that the rebate increases consumer's willingness to pay for any given quantity. Intuitively, because consumers are refunded a fraction t of their expenditure, they behave as if the market price were discounted by the factor $1 - t$.

- c. How does the monopolist's output respond to the tax-rebate program?

We now evaluate how monopolies respond when tax revenue is rebated to consumers. The monopolist continues to pay the revenue tax, so profits are given by:

$$\pi(x) = (1 - t)\left(\frac{v'(x)}{1 - t}\right)x - c(x) = v'(x)x - c(x)$$

This simplification reveals the central mechanism at work: the reduction in revenue due to taxation is exactly offset by the increase in consumer's willingness to pay generated by the rebate. To determine the optimal output, differentiate profits with respect to x :

$$\frac{d\pi}{dx} = \frac{d}{dx} [v'(x)x] - c'(x) = v'(x) + v''(x)x - c'(x)$$

The monopolist's first-order condition is therefore:

$$v'(x) + v''(x)x = c'(x) \quad \rightarrow \quad C1$$

Crucially, the tax rate t does not appear in C1. This condition coincides exactly with the monopolist's optimality condition in the absence of any tax, when inverse demand is simply $p(x) = v'(x)$.

The economic intuition is straightforward. The tax alone reduces marginal revenue and would tend to decrease output. However, rebating tax revenue proportionally to consumers increases inverse demand by precisely the same amount. Under quasilinear utility, these two effects cancel out perfectly.

As a result, the monopolist's equilibrium output is unchanged by the tax-rebate program.

QUESTION 12B7

Consider the widget market. The total demand by men for widgets is given by $x_m(p) = a - \theta_m p$, and the total demand by women is given by $x_w(p) = a - \theta_w p$, where $\theta_w < \theta_m$. The cost of production is c per widget.

- a. Suppose the widget market is competitive. Find the equilibrium and quantity sold

Men's and women's demands for widgets are $x_m(p) = a - mp$; $x_w(p) = a - wp$ $w < m$. Aggregate market demand is therefore:

$$X(p) = x_m(p) + x_w(p) = 2a - (m + w)p$$

Production occurs under constraint marginal cost c . In a perfectly competitive market, firms take prices as given and produce until price equals marginal cost. Hence the competitive equilibrium price is:

$$p^c = c$$

Substituting into demand, total quantity sold is:

$$X^c = 2a - (m + w)c$$

Individual quantities are:

$$X_m^c = a - mc; \quad X_w^c = a - wc$$

Therefore, competition delivers marginal-cost pricing and maximizes total surplus. Both men and women consume a positive quantity provided $a > mc$; since $w < m$ This condition guarantees positive demand for both groups.

- b. Suppose, instead, that firm A is a monopolist of widgets [also make this assumption in c and d] If firm A is prohibited from "discriminating" (i.e. charging different prices to men and women), what is its profit-maximizing price? Under what conditions do both men and women consume a positive level of widgets in this solution?

Firm A is now a monopolist and must charge a single uniform price to all consumers. As long as both men and women are served, the monopolist faces aggregate demand is:

$$X(p) = 2a - (m + w)p \rightarrow \text{with inverse demand} \rightarrow p(X) = \frac{2a - X}{m + w}$$

The monopolist chooses output X to maximize profit:

$$\pi(X) = p(X)X - cX = \frac{(2a - X)X}{m + w} - cX$$

The first-order condition is

$$\frac{d\pi}{dX} = \frac{2a - 2X}{m + w} - c = 0 \implies X^M = \frac{2a - c(m + w)}{2}$$

The corresponding monopoly price is:

$$p^M = \frac{2a - X^M}{m + w} = \frac{2a + c(m + w)}{2(m + w)}$$

To determine whether both groups consume, note that men have the steeper demand curve. Hence the binding condition is $xm = a - mp^M > 0$. Substituting for p^M gives:

$$a > \frac{m}{2(m + w)} [2a + c(m + w)]$$

If this condition fails, men are excluded and only women are served under uniform monopoly pricing.

- c. **If firm A has produced some total level of output X , what is the welfare-maximizing way to distribute it between the men and the women? (Assume here and below that Marshallian aggregate surplus is a valid measure of welfare).**

Suppose firm A has already produced total output X . The question is how this output should be distributed between men and women to maximize welfare, measured by Marshallian aggregate surplus. Total welfare is:

$$W = \int_0^{xm} (a - mq) dq + \int_0^{xw} (a - wq) dq$$

Subject to the feasibility constraint $xm + xw = X$. The welfare-maximizing allocation equates marginal willingness to pay across groups: $a - m \cdot xm = a - w \cdot xw$. This condition implies $m \cdot xm = w \cdot xw$. Using feasibility constraint, the efficient allocation is:

$$Xm = \frac{w}{m + w}X, \quad Xw = \frac{m}{m + w}X$$

Since $m > w$, men have a more price-sensitive demand curve, while women have a flatter (less price sensitive) demand curve. Because women's marginal willingness to pay falls more slowly with quantity, they must consume a larger quantity in order for marginal valuations to be equalized across groups. This allocation minimizes deadweight loss by ensuring that, at the margin, each unit is consumed by individuals who value it equally.

- d. Suppose that firm A is allowed to discriminate. What price does it charge? In this case where the nondiscriminatory solution in b has positive consumption of widgets by both men and women, does aggregate welfare as measured by Marshallian aggregate surplus rise or fall relative to when discrimination is allowed? Relate your conclusion to your answer in c. What if the nondiscriminatory solution in b has only one type of consumer being served?

Firm A is now allowed to charge different prices to men and women. The monopolist solves two independent profit maximization problems.

FOR MEN

$$\pi_m = (p_m - c)(a - mp_m) \rightarrow \text{first - order condition} \rightarrow p_m^D = \frac{a + mc}{2m}$$

FOR WOMEN

$$\pi_w = (p_w - c)(a - wp_w) \rightarrow \text{first - order condition} \rightarrow p_w^D = \frac{a + wc}{2w}$$

Since $w < m$, women are less price sensitive and are charged a higher price: $p_w^D > p_m^D$. That means that the monopolist charges a lower price to men and a higher price to women.

From part c, welfare maximization requires

$$\frac{x_m}{x_w} = \frac{w}{m}$$

So efficiency requires women to consume more than men. Under uniform monopoly pricing (part b), both groups face the same price p^M , implying:

$$\frac{x_m^M}{x_w^M} = \frac{a - mp^M}{a - wp^M}$$

Which generally differs from the efficient ratio w/m . In particular, uniform pricing tends to allocate relatively too much output to men (the more elastic group) compared to the efficient allocation.

Allowing price discrimination introduces a price differential with $p_m^D < p_w^D$. This increases women's consumption relative to men's and moves the allocation closer to the welfare maximising rule derived in part c. In addition, total output increases under discrimination. Both effects reduce deadweight loss, so Marshallian aggregate surplus rises relative to uniform monopoly pricing.

If under uniform monopoly pricing, only one group is served, then price discrimination eliminates the exclusion inefficiency by opening the second market. In this case, aggregate welfare strictly increases.

QUESTION 11D2

Consider a setting with I consumers. Each consumer i chooses an action $h_i \in \mathbb{R}_+$. Consumer i 's derived utility function over her choice of h and the choices of other consumers takes the form $\phi_i(h_i, \Sigma_{j \neq i} h_j) + w_i$ where $\phi_i(\cdot)$ is strictly concave. Characterize the optimal levels of $h_1, \dots, h_I$. Compare these with the equilibrium levels.

What tax/subsidy scheme induces the optimal outcome?

We consider an economy populated by I consumers, indexed by $i = 1, \dots, I$. Each consumer chooses a scalar action $h_i \in \mathbb{R}$. The interpretation of h_i is deliberately general: it may represent consumption of a good, effort, participation in an activity, or any individual decision whose consequences extend beyond the decision maker.

Consumers i 's preferences are represented by a quasilinear utility function of the form:

$$U_i(h_i, h_{-i}, w_i) = u_i(h_i, h_{-i}) + w_i$$

Where h_{-i} represents the vector of actions chosen by all consumers other than i , and w_i is wealth. The function $u_i(\cdot)$ captures all non numerative aspects of utility and is assumed to be continuously differentiable and strictly concave in h_i .

This structure is important because it implies that marginal utility of wealth is constant and normalised to one, leading us to analyze the choice of h_i independently of income effects.

Lastly, we allow each agent's action to directly affect the utility of others, so for some $i \neq j$:

$$\frac{\partial u_j}{\partial h_i} \neq 0$$

Social optimum

A social planner chooses the vector $(h_1, \dots, h_I)$ to maximize aggregate welfare, defined as the sum of individual utilities:

$$\max_{h_1, \dots, h_I} \sum_{i=1}^I u_i(h_i, h_{-i})$$

Because preferences are quasilinear, wealth terms drop out of the maximization problem and don't play a role in determining efficiency. To characterize the optimal allocation, consider a marginal increase in h_i . The effect of this change on total welfare is given by:

$$\max_{h_1, \dots, h_I} \sum_{j=1}^I u_j(h_i, h_{-j}) = \sum_{j=1}^I \frac{\partial u_j}{\partial h_i} \xrightarrow{\text{satisfies}} \sum_{j=1}^I \frac{\partial u_j}{\partial h_i} = 0, \quad \forall i$$

$$(1) \quad \frac{\partial u_i}{\partial h_i} + \sum_{j \neq i} \frac{\partial u_j}{\partial h_i} = 0$$

The first term represents the marginal private benefit of increasing h_i , while the second term captures the marginal external effect of h_i on all other consumers. At the social optimum, the planner equates the total marginal effect of each action to zero.

Decentralized equilibrium

In the decentralized economy, each consumer chooses their action independently, taking the actions of others as given. Consumer i solves:

$$\max_{h_i} u_i(h_i, h_{-i}) + w_i$$

Since wealth enters linearly, it does not affect the marginal incentives governing the choice of h_i . The first-order condition for an interior optimum is therefore:

$$\frac{\delta u_i(h_i, h_{-i})}{\delta h_i} = 0 \quad \text{for all } i$$

Let the vector $(h'_1, \dots, h'_I)$ denote the Nash equilibrium defined by the system of equations. Each agent chooses h_i so as to equate their private marginal benefit to zero, without accounting for the effects of their action on the utilities of others.

Comparison

Comparing the conditions of the social optimum and the decentralized equilibrium highlights the source of inefficiency. The Nash equilibrium condition is $\frac{\delta u_i}{\delta h_i} = 0$, whereas the socially

optimal condition is $\frac{\delta u_i}{\delta h_i} + \sum_{j \neq i} \frac{\delta u_j}{\delta h_i} = 0$. The discrepancy between these two conditions is

given by the marginal external effect $\sum_{j \neq i} \frac{\delta u_j}{\delta h_i}$

- If this term is negative $\rightarrow$ each agent imposes a negative externality on others and the decentralized equilibrium involves excessive levels of h_i relative to social optimum.
- If term is positive $\rightarrow$ agents generate positive externalities and equilibrium levels of h_i are inefficiently low.

In general the decentralized equilibrium fails to internalize the full social consequences of individual actions and is therefore inefficient.

Therefore, to decentralize the socially optimal allocation, suppose the government introduces a linear tax or subsidy on each action h_i at rate τ_i . Consumer i 's problem becomes:

$$\max_{h_i} u_i(h_i, h_{-i}) - \tau_i h_i + w_i \rightarrow \text{first order condition} \rightarrow \frac{\delta u_i}{\delta h_i} - \tau_i = 0$$

To introduce the socially optimal choice h_i^* , the tax rate must be chosen so that this condition coincides with the social planner's condition (1). This requires:

$$\tau_i = - \sum_{j \neq i} \frac{\delta u_j(h')}{\delta h_i}$$

Therefore, each consumer faces a tax equal to the marginal external effect of their action on others. This Pigouvian tax scheme aligns private incentives with social welfare and ensures that the decentralized equilibrium coincides with the socially optimal allocation.